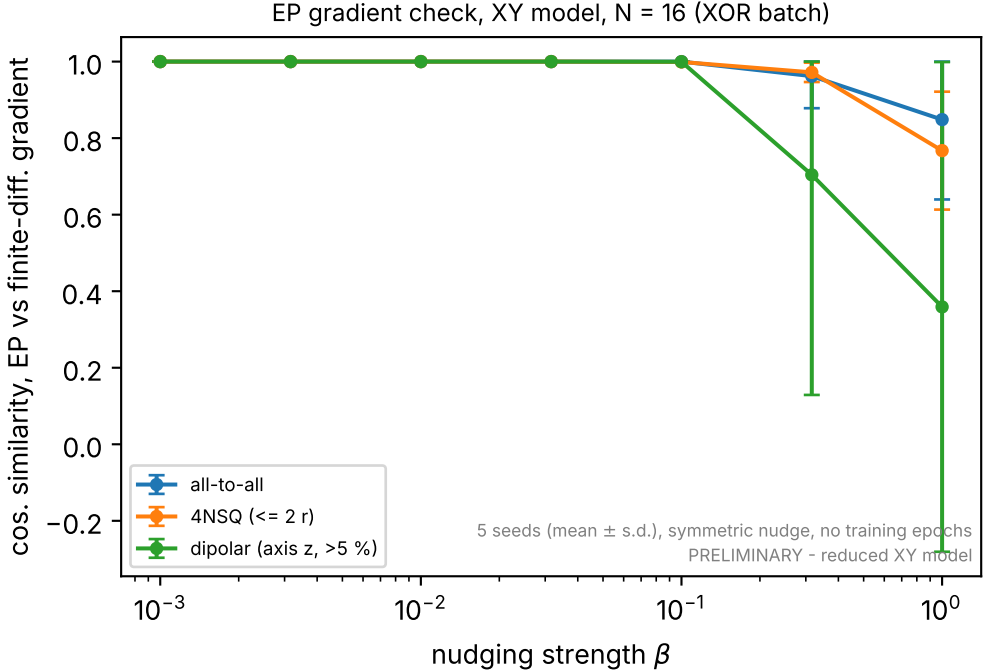
EP gradient check, XY model, N = 16 (XOR batch)
cos. similarity, EP vs finite-diff. gradient
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
all-to-all
4NSQ (<= 2 r)
dipolar (axis z, >5 %)
5 seeds (mean ± s.d.), symmetric nudge, no training epochs
PRELIMINARY - reduced XY model
$10^{-3}$
$10^{-2}$
$10^{-1}$
$10^{0}$
nudging strength $\beta$